Huan Nguyen

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EDUCATION

UC Berkeley **Expected: ~Spring 2027**

Bachelor of Science, EECS *Berkeley, CA*

- **Activities**: Solar Vehicle Team's Electrical Engineer, Unity Theme Program Member, Formula Electric Berkeley's General Member.
- **Completed Relevant Coursework**: IC Devices, Signals and Systems, Data Structures, Machine Structures, Microelectronic Devices and Circuits, Photovoltaic Devices, Microfabrication Technology.

SJCC/EVC **Expected: ~Summer 2026**

Associate in Arts, Economics *San Jose, CA*

WORK EXPERIENCE

Calsol Solar Vehicle Team **January 2026 – Present**

Project Lead *Berkeley, CA*

- Leading the design and development of an MPPT Interface PCB to integrate solar maximum power point tracking controllers into the vehicle's electrical system.
- Develop and support embedded firmware in C, applying knowledge of CAN bus, GPIO, and communication protocols for vehicle electronics.

Boys & Girls Club **April 2025 – August 2025**

Youth Development Professional *San Jose, CA*

- Designed and led an independent STEM class for the entire summer, managing curriculum planning, instruction, and classroom engagement.

Dell Technologies **October 2019 – June 2023**

Business Intern *San Jose, CA*

- Participated in a mentorship experience focused on understanding Dell Technologies' business models and organizational structure. Engaged with insights from various industry professionals.

PROJECTS

MPPT Interface PCB (Calsol Solar Vehicle Team) [Nov 2026 - In Progress]

- Designing a CAN communication interface to bridge MPPT controllers operating at 500 kbps with the vehicle's 1 Mbps CAN bus using an ESP32-based system. Developing schematic designs incorporating CAN transceivers, voltage regulation, and 24V-to-12V DC-DC power conversion for compatibility with the vehicle's electrical architecture.

NMOS Integrated Circuit Fabrication & Characterization [Jan 2026 - Apr 2026]

- Fabricated and electrically characterized NMOS devices on a p-type silicon wafer in a semiconductor cleanroom using photolithography, oxidation, wet etching, doping/diffusion, polysilicon deposition, and aluminum metallization.

NMOSFET Design Project [Dec 2025 - Dec 2025]

- Designed and simulated a 25 nm gate-length NMOS transistor using Synopsys Sentaurus TCAD as part of UC Berkeley's EE130 (Integrated-Circuit Devices) course.

SKILLS

- **Skills:** Microfabrication, PCB Design, LTSpice, Computer Simulations, Python, C++, Java, Assembly, etc.